REVIEWS

Superconductivity without phonons

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The idea of superconductivity without the mediating role of lattice vibrations (phonons) has a long history. It was realized soon after the publication of the Bardeen–Cooper–Schrieffer (BCS) theory of superconductivity 50 years ago that a full treatment of both the charge and spin degrees of freedom of the electron predicts the existence of attractive components of the effective interaction between electrons even in the absence of lattice vibrations—a particular example is the effective interaction that depends on the relative spins of the electrons. Such attraction without phonons can lead to electronic pairing and to unconventional forms of superconductivity that can be much more sensitive than traditional (BCS) superconductivity to the precise details of the crystal structure and to the electronic and magnetic properties of a material.

Superconductivity, the dissipationless flow of electrical current, is a striking manifestation of a subtle form of quantum rigidity on the macroscopic scale. In such a quantum state, the low-energy behaviour is insensitive to many perturbations, and is typically characterized by a small number of parameters that can be determined experimentally but are difficult to calculate accurately from first principles[1]. The major achievement of Bardeen, Cooper and Schrieffer was to identify the fundamental organizing principle responsible for superconductivity, and show in a simple model calculation how the low-lying excited states could be characterized by three macroscopic parameters that were easily determined from experiment[2,3].

In real metals at temperatures that are low, but above the superconducting state, the experimental observations can be accounted for in terms of a low density gas of thermally excited particles (and antiparticles) with the same charge and spin, but not the same mass, as the electron (and positron). These particles are referred to as quasiparticles. In the hypothetical absence of the electron–electron and electron–ion interactions, these quasiparticles reduce to bare electrons. In real metals, one may think of a quasiparticle as an electron plus a co-moving screening cloud. In extreme cases, such as the heavy electron superconductors that are of particular interest in this Review, the quasiparticles can behave as if they have, for example, effective masses two or more orders of magnitude greater than that of electrons and hence move relatively slowly[4–8]. The low velocity of the quasiparticles reflects the fact that the transport of charge in these systems is strongly impeded by the lattice potential and collisions with other particles. We note that the quasiparticle picture described here appears to have a wide range of applicability—remarkably, even on the border of, but not too close to, instabilities.

At a deep level, the electron–electron interaction is an induced interaction: that is, one electron (the polarizer) induces a change in the electromagnetic field in free space and the second electron (the analyser) samples this change. One can also think of the interaction between quasiparticles in the same manner, but the electromagnetic field now also includes that produced by the other particles in the material. We refer to the total electromagnetic field seen by the quasiparticles as the effective field. This suggests that the quasiparticle interaction can be very different from that of electrons, and is the key to understanding superconductivity. In particular, the charge–charge interaction between quasiparticles can have attractive regions in space owing to the collective effects of the electron and lattice system. Importantly, the quasiparticles also experience an effective spin–spin interaction (to be discussed below) that can be several orders of magnitude stronger than the conventional magnetic dipole interaction between bare electrons.

Where the quasiparticle interaction has attractive regions, a new kind of quantum order can emerge in which the quasiparticles form pairs locked into a coherent state extending over macroscopic dimensions. This is the phenomenon of superconductivity.

In conventional superconductors (for example, Al, Pb and Hg), the effective charge–charge interaction between quasiparticles is dominant. The attractive part of the interaction in these cases arises from the induced effects of quantized lattice vibrations or phonons. In this Review, we begin with a primer on electron–phonon-mediated superconductivity and then explain how quasiparticle pairs can form via the above mentioned spin–spin interaction that does not involve phonons explicitly and depends on the spins of the interacting quasiparticles rather than their charges.

Phonon-mediated superconductivity

For the first 40 years after the discovery of superconductivity in 1911 (ref. 9), those seeking a microscopic mechanism for superconductivity did not consider the possibility that phonons could play any part. Rather, the focus was on a possible special role for the crystal structure of the lattice[10] or on the possibility that, if one treated properly the consequences of the Coulomb interaction between the electrons[11,12], superconductivity might emerge. Matters changed when researchers working at Rutgers University and the National Bureau of Standards discovered the isotope effect—that in lead, the transition temperature below which superconductivity was found depended inversely on the square root of the isotopic mass[13,14]. This discovery motivated Frohlich[15] and Bardeen[16] to explore the possibility that it was the change in the quasiparticle energy spectrum due to interactions with the phonons that led to superconductivity, an approach that soon turned out be to fruitless.

Frohlich had also suggested[17] that somehow the attractive part of the phonon-induced interaction between quasiparticles might play a part. However, this proposal also seemed flawed, as it appeared that the much stronger repulsive direct Coulomb interaction would swamp the effect. (Evgeny Lif'shitz mentioned to one of us that Landau said at one point "You can't repeal Coulomb's law".) This objection turned out not to be insuperable, as a few years later, Bardeen and one of us[18] were able to show, in a simple model

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calculation, that when full account was taken of screening[19] by the electron and lattice system as a whole one could still get a regime in space and time in which the interaction between quasiparticles is attractive.

Cooper[20] then showed how this effective interaction between quasiparticles could give rise to the formation of quasiparticle pairs (see Fig. 1a for a real space illustration of an *s*-wave Cooper-pair state). Within a year, Bardeen, Cooper and Schrieffer developed a simple microscopic description of the superconducting ground state in terms of Cooper pairs.

An effective attractive component of the overall interaction between quasiparticles was thereby established as the key to a superconducting transition. For the next nearly 30 years, the primacy of the phonon-induced attraction—with a predicted ceiling to the superconducting transition temperature (T_c) below the boiling point of liquid nitrogen—appeared to be well established. Matters changed with the discovery by Bednorz and Müller[21] of the first member of the copper oxide family of superconductors, which have T_c values of up to 160 K (twice the boiling point of liquid nitrogen). These unexpectedly high T_c values and other anomalous properties of the copper oxides led theorists to explore in detail the possibility of superconductivity without phonons, that is, of an effective attractive interaction between quasiparticles that does not originate in their coupling to lattice vibrations.

This search had begun earlier, in particular with the suggestion that a spin–spin quasiparticle interaction[22], analogous to that thought to stabilize superfluidity in liquid ^{3}He (refs 23, 24), might also be relevant to superconductivity on the border of itinerant-electron magnetism[25], where the spin–spin interaction described in more detail in the next section is expected to be important. Superconductivity on the border of ferromagnetism[25] has proved to be an elusive phenomenon, and has been observed only relatively recently following a long search[26,27]. However, for nearly a decade before the Bednorz and Mueller discovery, evidence had been accumulating for superconductivity on the border of antiferromagnetism in low-dimensional organic metals[28] and a family of 'heavy electron' compounds[4–7,29–32].

Whereas these materials on the boundary of antiferromagnetism superconduct only at comparatively low temperatures, both their normal state and superconducting properties seemed quite different from those of the conventional metallic superconductors. This led to speculations that the spin–spin quasiparticle interaction, extended from the case of a nearly ferromagnetic metal to the seemingly more subtle case of a nearly antiferromagnetic system, might be the dominant mechanism for quasiparticle pairing in these materials[33–35]. It is this possibility for superconductivity without phonons, both in the nearly ferromagnetic and the nearly antiferromagnetic metals, that we consider in this Review.

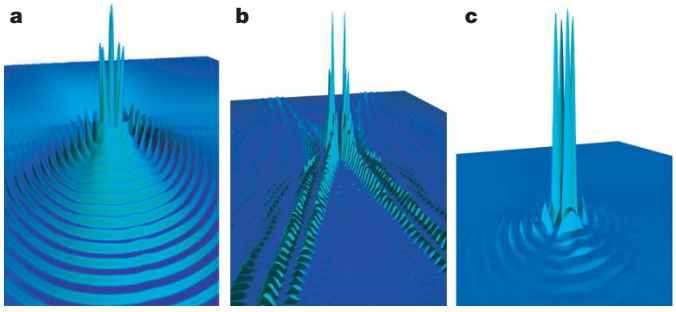


Figure 1 | Cooper-pair states in real space in two dimensions. a–c, The probability of finding one quasiparticle in a Cooper-pair state given that the other partner is at the origin. The symmetric form in **a** is characteristic of an *s*-wave spin-singlet state. The two-fold symmetry in **b** is characteristic of one of the possible *p*-wave spin-triplet states, whereas the four-fold symmetry in **c** is characteristic of a *d*-wave spin-singlet state. The Cooper-pair states shown are predicted for the model interaction described in the text just below the superconducting transition temperature, T_c. In **a** we have used a simple model of the phonon-mediated interaction[3] and in **b** and **c** realistic models of the electronic structure of the ruthenates (**b**) and the copper oxides (**c**)[51]. In the *s*-wave and *p*-wave cases (**a** and **b**, respectively), the probability distribution function extends over many lattice sites, which is the signature of a weakly bound large-sized Cooper pair reflecting a low T_c. In contrast, in the *d*-wave case (**c**), the probability distribution is strongly peaked at the nearest neighbour lattice site to the origin and quickly decays with distance. This indicates a small Cooper-pair size and correspondingly high T_c.

Density and magnetically mediated superconductivity

In this section, we describe a simple framework for thinking about quasiparticle interactions of which the phonon-mediated interaction is a special case. We consider the case in which a quasiparticle at some point $\mathbf{r}$ and time t would see two effective fields, one proportional to the charge density $n(\mathbf{r},t)$ coupled to the quasiparticle charge e, and the other proportional to the spin density or magnetization $\mathbf{m}(\mathbf{r},t)$ coupled to the quasiparticle spin $\mathbf{s}$. In the case of a bare electron, these fields would reduce to the electric and magnetic fields, respectively, that obey Maxwell's equations. In principle, there will also be analogous fields coupled to the quasiparticle current and other quasiparticle properties that will not be considered further here[36,37]. We implicitly assume that the mean value of the density has been subtracted out, and that the system is not magnetically ordered.

In this phenomenological model, the interaction of the given quasiparticle with the fields can be expressed in the form

$$-e[g_n\, n(\mathbf{r},t)] - \mathbf{s}\bullet[g_m\, \mathbf{m}(\mathbf{r},t)] \qquad (1)$$

where g_n and g_m are empirically determined parameters that measure the strength of the coupling to the charge and to the spin, respectively. Note that in electromagnetism the relevant coupling parameters are also obtained experimentally. More generally, the fields seen by the quasiparticles may depend not only on the local densities, but also on the densities in the neighbourhood and at slightly earlier times. Mathematically, the above relation is then replaced by a convolution in space and time. An example of this is the polarization potential approach introduced by one of us for the helium liquids[38].

In quantum electrodynamics, the effective electron–electron interaction including quantum corrections can be calculated by averaging over the statistical distribution of the Maxwell field. In an analogous way, one can obtain the low-frequency quasiparticle–quasiparticle interaction by averaging over the appropriate joint statistical distribution of effective fields. This distribution is likely to be very complex, and thus the usefulness of the effective field picture rests on the fact that it works—namely, that simple approximations can yield significant physical insights into the behaviour of correlated quantum matter.

An intuitive picture of the effective interaction starting with the spin–spin interaction can be given as follows. The spin $\mathbf{s}'$, the polarizer, generates a bare interaction field, taken for simplicity to be localized at the origin in space and at the origin in time, of strength $g_m\mathbf{s}'$. In the linear response approximation, this field induces a magnetization given by $\mathbf{m}(\mathbf{r},t) = g_m\mathbf{s}'\chi_m(\mathbf{r},t)$,where χ_m is the non-local magnetic susceptibility. This induced magnetization produces an induced effective field $g_m\mathbf{m}(\mathbf{r},t)$ that acts on spin $\mathbf{s}'$, the analyser. The induced interaction takes the familiar form of a magnetic moment in an external field given by the second term in equation (1), which in this case reduces to $-g_m{}^2\mathbf{s}\bullet\mathbf{s}'\chi_m(\mathbf{r},t)$. A similar argument applied to the charge–charge interaction yields an induced interaction of the form $-ee'g_n{}^2\chi_n(\mathbf{r},t)$, where χ_n is the non-local charge density susceptibility and $e = e'$ is the charge of the interacting quasiparticles. This leads to an overall induced interaction:

$$V_{\mathrm{ind}}(\mathbf{r},t) = -\,e\,e'\,g_n{}^2\,\chi_n(\mathbf{r},t) - \mathbf{s}\bullet\mathbf{s}'\,g_m{}^2\,\chi_m(\mathbf{r},t) \qquad (2)$$

In our model, this induced interaction is in addition to a residual short-range repulsion that prevents charge carriers from coming to

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the same place at the same time. An extension of this treatment to the 'strong coupling' limit, where the effective fields are not weak, is discussed below.

For comparison between the predictions of the model interaction (equation (2)) and experiment, it is best to use as inputs the experimentally determined susceptibilities χ_n and χ_m and molecular field constants g_n and g_m. To explore in a qualitative way the range of possible behaviours, it is fruitful, however, to consider simple theoretical models for the susceptibilities in equation (2).

To a first approximation, χ_n and χ_m could be replaced by the susceptibility of non-interacting electrons in a periodic crystal potential, namely, the Lindhard function, which can be obtained by means of standard energy band theory. Although this approximation leads one to understand how the effective interaction between quasiparticles can have attractive regions in space and time, even when phonons are not included, it normally leads to values of T_c so low as to make the experimental observation practically impossible[11,12,39].

The values of T_c could potentially rise to the experimentally accessible range if the density or magnetic susceptibilities χ_n and χ_m (and hence the attractive parts in equation (2)) were enhanced over that of the Lindhard function. Such considerations would naturally lead one to look for superconductivity on the border of density or magnetic instabilities. An improved representation of χ_n and χ_m could be obtained in the random phase approximation[19], a generalization of the Weiss-Stoner model for the magnetic susceptibility, which yields χ_n and χ_m in terms of the Lindhard functions and the effective field constants g_n and g_m. A still better approximation is obtained when these starting effective field constants are replaced by effective temperature dependent ones determined by a self-consistency requirement[40,41].

In contrast to the random phase approximation, this latter self-consistent renormalization model yields values of ferromagnetic or antiferromagnetic transitions that are generally in good agreement with experiment[42]. In particular, this model is in keeping with the suppression of long-range magnetic order at finite temperatures expected for systems in one or two dimensions.

Apart from the molecular field constants, the model interaction in the self-consistent renormalization approximation is then completely specified by the underlying energy band structure that defines the Lindhard functions.

Where to look

The problem of quasiparticle pairing via equation (2) is made subtle by the fact that the interaction will in general have both attractive and repulsive regions in space and time. The issue is then whether a Cooper-pair wavefunction can be constructed that has a large amplitude in the space-time regions where the interaction is attractive and a small amplitude elsewhere. An additional complication compared to the more conventional particle pairing in free space, where all momentum states are available for optimizing the pair waveforms, is the following. In free space, the low-energy electron and positron excitations have momentum close to zero. In the case of a finite density of electrons, the corresponding low energy particle and antiparticle states have a finite momentum. The momenta of zero energy excitations determine a surface in momentum space referred to as the Fermi surface. In the present case, the available states for the pair states are generally restricted to the neighbourhood of the Fermi surface. This implies that the pair wavefunction oscillates in space with wavevector connected with the diameter of the Fermi surface. Because the interaction is dynamical, the appropriate Cooper state will also exhibit a non-trivial time dependence.

In order for such an interaction to bring about quasiparticle pairing, the allowed oscillatory nature of the Cooper state, restricted by the Fermi surface, must match that of the interaction.

In simple metals, the effective charge–charge interaction is dominant; it is repulsive at the origin ($\mathbf{r} = t = 0$) and becomes attractive only at finite times ($\mathbf{r} = 0$, $t > 0$) owing to the effects of phonons (Fig. 2a, b). This then leads to a Cooper state with maximum amplitude at finite time and with zero relative angular momentum (*s*-wave, Fig. 1a). The Pauli principle requires that the spins of the paired quasiparticles be in the singlet state. This time-delayed charge–charge interaction is the key to understanding the conventional superconductors, but may only be part of the story in strongly correlated electron systems.

Particularly in systems close to magnetic instabilities (for example, close to ferromagnetism or antiferromagnetism), the spin susceptibility $\chi_m(\mathbf{r},t)$ can be appreciable, and thus we expect the spin–spin interaction (the second term in equation (2) to be important. A qualitatively new feature of this interaction is that contrary to the charge–charge interaction where $ee' = e^2$ is always positive, the inner products of the two spins can be positive or negative depending on the relative spin orientations of the interacting quasiparticles.

On the border of a ferromagnetic instability, $\chi_m(\mathbf{r},t)$ is maximum and positive at the origin and decays slowly in space and time. Therefore, the spin interaction is attractive for parallel spins, that is, for the spin-triplet state (Fig. 1c). The Pauli principle then requires the Cooper state to have odd angular momentum. Generally, the attraction is strongest in the lowest allowed angular momentum state, or the *p*-wave state (Fig. 1b).

On the other hand, on the border of antiferromagnetism, $\chi_m(\mathbf{r},t)$ is also maximum and positive at the origin and tends to decay in time, but more crucially it oscillates in space with a period comparable to the lattice spacing (Figs 2d, 3). For either the spin-singlet or spin-triplet states, this interaction will have both repulsive and attractive regions in space that tend to produce cancelling effects. It is still possible to get a superconducting instability, provided that the spatial

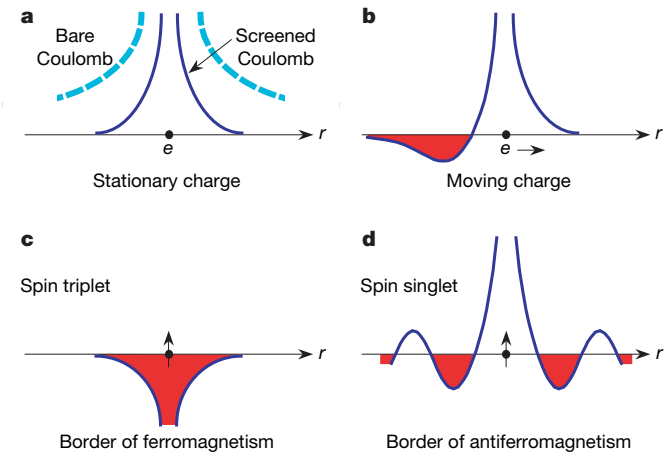


Figure 2 | Schematic illustration of the charge-charge and spin-spin quasiparticle interaction in a metal. Panels **a** and **b** depict the charge–charge interaction potential versus distance (r) produced by a test charge at rest in **a** and moving to the right in **b**. The dashed curve in **a** represents the static Coulomb repulsion and the solid line represents this interaction as screened by the effects of other charged particles (ions and electrons) in the material. We note that in this simple model the interaction is everywhere repulsive in the static limit. When the test charge is moving, however, the interaction develops an attractive region directly behind it (**b**). This attraction is caused by the polarization of the positively charged and heavy ions that lie behind the moving charge and its electron screening cloud. A similar phenomenon arises in the classical description of an object moving through a medium. For example, a fast moving car creates a low pressure (attractive) region just behind it. Panels **c** and **d** represent effective spin–spin interaction potentials set up by a test spin at rest in a material on the border of long-range magnetic order. The interaction potential shown is attractive for two quasiparticles with the same spin (spin triplet state) on the border of ferromagnetism (**c**) and is oscillatory in space on the border of antiferromagnetism (**d**). The illustration in **d** is for the case where the two interacting quasiparticles have opposite spins (spin singlet state). In real materials, the interaction potential is expected to display a complex oscillatory pattern both on the border of ferromagnetism and antiferromagnetism.

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variation of the Cooper state can be properly adjusted to match the oscillations of the spin interaction. Generally, the instability is found in the spin-singlet state, which must necessarily have even angular momentum. As the interaction is repulsive at the origin, a non-zero angular momentum state, typically a *d*-wave state, is favoured (Fig. 1c).

We note that the analogues of the ferromagnetic and antiferromagnetic instabilities can exist in the density (described by $\chi_n(\mathbf{r},\mathbf{t})$) as well[43]. The pattern of oscillations in $\chi_n(\mathbf{r},t)$ is determined by the wavevector at which the density response is maximum, and this too can lead to unconventional pairing states. In contrast to the magnetic interaction where the sign of the interaction is different for spin-triplet and spin-singlet Cooper pairs, the density interaction is oblivious to the spin state of the Cooper pairs. Hence, the magnetic interaction offers more freedom to match the attractive regions of the oscillations of the interactions with the quasiparticle states near the Fermi surface available to construct a Cooper-pair wavefunction.

Some surprises

On the border of ferromagnetism the induced interaction is purely attractive at short distances, whereas on the border of antiferromagnetism it is likely to have repulsive regions when the two interacting quasiparticles are close to each other. It might therefore be expected that the search for exotic pairing states on the former border would have proved the more fruitful. This seemed to have been borne out by the discovery of the superfluidity of liquid ^{3}He in which the uniform magnetic susceptibility is strongly enhanced[44]. Given the abundance of metals that exhibit strong ferromagnetic correlations, it is more than a little surprising that it took a quarter of a century to discover a superconducting analogue of liquid ^{3}He, namely, the layered perovskite Sr_2RuO_4 that has a T_c value two orders of magnitude below that of the high T_c copper oxides[26].

Even more perplexing on the other hand is the fact that many examples of superconductivity on the border of antiferromagnetism have been found in the intervening period[4–7,28–32]. These findings would suggest at first sight that the magnetic interaction is giving us little or no insight on where to look for exotic forms of superconductivity.

However, the great lesson of the past decade is that subtleties in the magnetic interaction model only come to the surface after a careful examination of its properties. Although the idea of a magnetic interaction goes back nearly half a century, the computer algorithms and hardware necessary for an exploration of the detailed predictions of the model have only become available more recently.

The results of these theoretical investigations have led to an intuitive understanding of the following: (1) why superconductivity can be particularly robust on the border of antiferromagnetism in a quasi-two-dimensional tetragonal system with high characteristic spin fluctuation frequencies[30,45–48]; (2) how the charge–charge and spin–spin interactions can in some cases work coherently to stabilize anisotropic Cooper-pair states; and (3) why pairing on the border of ferromagnetism is hampered by quite a number of effects, and may depend on subtle details of the electronic structure, that is, features (absent in liquid ^{3}He) of the energy band of the periodic crystal potential[49–51]. Illustrations of these ideas are given below.

The first reason for the robustness of pairing in the presence of antiferromagnetic correlations in a tetragonal structure is that the amplitude of the oscillations in the interaction is strong because of the low dimensionality. The energy density of the interaction waves created by the polarizer falls off more gradually in two dimensions (as 1/distance) than in three dimensions (as 1/distance^2). The second reason is that the repulsive regions of the interaction in real space are along the diagonals of the lattice given that one quasiparticle is at the origin (see Fig. 3). In this case, the crystal symmetry allows one to choose a *d*-wave Cooper state with nodes along the diagonals, thereby neutralizing most of the repulsive regions while retaining the attractive regions. One can easily imagine that it will not always be possible to choose a Cooper-pair state in such an optimal way, and that the initial impression that the oscillations of the interaction are detrimental to superconductivity may only be wrong in special cases. In particular, as the tetragonal structure becomes more and more isotropic under otherwise similar conditions, the model predicts a decrease in the robustness of the pairing. The range in temperature and pressure over which superconductivity is observed was increased by about one order of magnitude in going from cubic $CeIn_3$ (refs 30, 52) to its tetragonal analogues $CeMIn_5$, where M stands for Rh, Ir or Co (refs 31, 32, 53–56; Fig. 4), as anticipated by the magnetic interaction model.

Another case where subtle features of the model considered here could explain puzzling superconducting properties is the first of the heavy-fermion superconductors, $CeCu_2Si_2$ (ref. 4), and the related

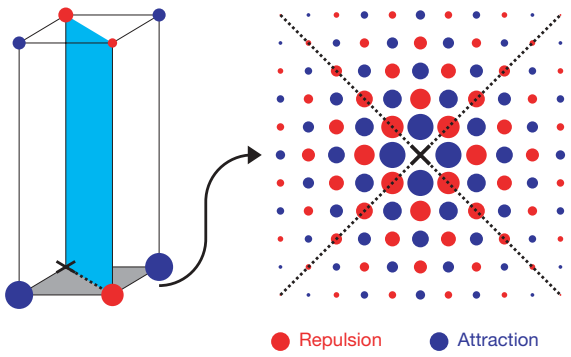


Figure 3 | Magnetic interaction potential in a lattice. Graphical representation of the static magnetic interaction potential in real space seen by a quasiparticle moving on a square crystal lattice given that the other quasiparticle is at the origin (denoted by a cross). The spins of the interacting quasiparticles are taken to be antiparallel, such that the total spin of the Cooper pair is zero. The dashed lines show the regions where the *d*-wave Cooper-pair state has vanishing amplitude. This is the state that best matches the oscillations of the potential, in that a quasiparticle has minimal probability of being on lattice sites when the potential induced by the quasiparticle at the origin is repulsive. The size of the circle in each lattice site is a representation of the absolute magnitude of the potential (on a logarithmic scale). This picture is appropriate for a system on the border of antiferromagnetism in which the period of the real space oscillations of the potential is precisely commensurate with the lattice.

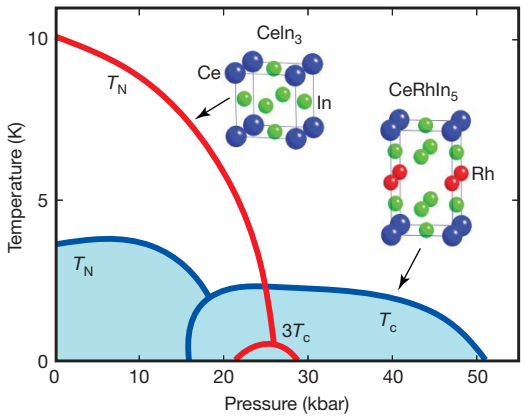


Figure 4 | Effect of electronic anisotropy. The schematic temperature–pressure phase diagram of two related heavy fermion compounds, $CeIn_3$ (refs 30, 52) and $CeRhIn_5$ (refs 53–55). These two materials differ in particular in the degree of anisotropy of the low energy excitation spectrum. As one would expect, the thermal fluctuations in the local magnetization lead to a smaller value of the magnetic transition temperature (Néel temperature, T_N) in the anisotropic material. By contrast, perhaps unexpectedly, T_c is greatly suppressed in the isotropic compound $CeIn_3$ (red lines) compared with $CeRhIn_5$ (blue lines). Both of these features are in qualitative agreement with the magnetic interaction model.

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compound, $CeNi_2Ge_2$ (refs 57–59). The appearance of two superconducting domes in the temperature–pressure phase diagram of $CeNi_2Ge_2$ (refs 58, 59) led some of us to suspect a different origin of superconductivity in the two cases, that is, magnetically mediated on the border of antiferromagnetism at low pressures and density-mediated on the border of an α–γ density instability at higher pressures. Such magnetic and density instabilities have been identified in $CeCu_2Si_2$, in which the two superconducting domes coalesce in stoichiometric samples but become separate and distinct in Ge doped samples[60,61] (Fig. 5). Our model quasiparticle interaction can naturally account for this behaviour.

In our model, the overall scale for T_c depends on the characteristic width of the energy band out of which the paired quasiparticles emerge. This suggests that a further increase in T_c could be obtained if the 4*f* Ce band were replaced, for example, by a wider 5*f* band under otherwise equivalent conditions, for example, in the same tetragonal structure and without moving too far from the border of the magnetic instability. This is indeed what happens when Ce in $CeMIn_5$ is replaced by the 5*f* metal Pu to obtain the related systems $PuMGa_5$, which leads to an increase of T_c by about another order of magnitude[62,63].

These aspects of our model can be understood even with a simple Lindhard function that represents real-space oscillations with a single wavelength of the order of the lattice spacing, as is to be expected for a system on the border of commensurate antiferromagnetism. A whole new level of complexity and subtlety arises when Lindhard functions for realistic energy bands are used (Fig. 6). The oscillation pattern in real space is a superposition of components of different wavelengths, and can vary dramatically with the filling level of the relevant energy band.

These rich oscillation patterns have proved to be a critical element in developing a qualitative understanding of the scarcity of examples of superconductivity on the border of ferromagnetism[51]. Another crucial factor in the case of the border of ferromagnetism is that the magnitude of the quantum mechanical average of the inner product of the two quasiparticle spins is a factor of three smaller in the spin-triplet state than in the spin-singlet state. This unique feature of the magnetic interaction means that pairing in the spin-triplet state has an inherent disadvantage relative to pairing in the spin-singlet state. The factor of three in the relative ineffectiveness of the former pairing is a property of a quantum spin 1/2, and does not arise for a classical spin or in an anisotropic system in which only fluctuations in the magnetization along a particular axis in the crystal are relevant. This suggests that pairing on the border of ferromagnetism is more likely to arise in systems with a strong magnetic anisotropy[49]. This prediction motivated the search for superconductivity on the border of ferromagnetism in UGe_2, which is known to have a strong uniaxial magnetic anisotropy. The search proved fruitful, for it led to the discovery of the first example of the co-existence of superconductivity and itinerant-electron ferromagnetism[27,64–67] (Fig. 7). Soon after, superconductivity was discovered in the itinerant-electron ferromagnets URhGe (ref. 68) and UIr (ref. 69) and even in the ε-phase of elemental Fe (ref. 70).

The magnetic energy scales in *d* metals may be much larger than in the *f* metals and one might expect, on the basis of the magnetic interaction model, that the spin-triplet *p*-wave superconductivity in the nearly ferromagnetic tetragonal compound Sr_2RuO_4 would occur at relatively high temperatures, despite the inherent disadvantage of spin-triplet pairing discussed above. That this is not the case may be understood in terms of the subtle oscillation pattern of the Lindhard function for a realistic model of the 'd_{xy}' energy band of Sr_2RuO_4 thought to be dominant for superconductivity. In particular, the form of the Fermi surface that arises in the presence of a periodic lattice potential tends to have nesting features that lead to real-space oscillations in the magnetic interaction that interfere with the long-wavelength components associated with ferromagnetic correlations. An additional detrimental factor to *p*-wave pairing in Sr_2RuO_4 arises from the fact that strong ferromagnetic correlations only occur when the Fermi surface is close to a 'van Hove' singularity point. The Fourier transform of the *p*-wave Cooper state vanishes at these points, and hence the superconducting energy gap is small in regions of high density of electronic states on the Fermi surface—that

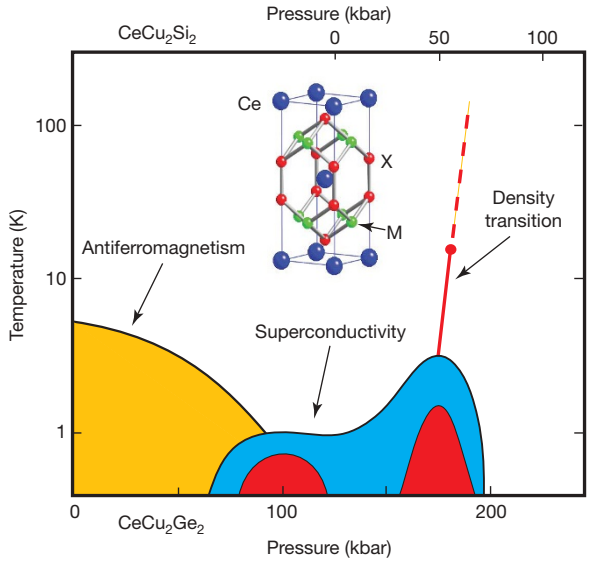


Figure 5 | Proximity to antiferromagnetic and density instabilities. Schematic temperature–pressure phase diagram for the class of compounds CeM_2X_2 having the crystal structure shown in the inset. In $CePd_2Si_2$ (ref. 30; that is, M = Pd, X = Si) and $CeRh_2Si_2$ (ref. 32), one only observes a superconducting dome on the border of metallic antiferromagnetism (left hand side of diagram), whereas in $CeNi_2Ge_2$ (refs 57–59) and in the close relatives $CeCu_2Si_2$ and $CeCu_2Ge_2$ (refs 4, 60, 61) one observes two superconducting domes, one of which is similar to that found in $CePd_2Si_2$ and $CeRh_2Si_2$. In $CeCu_2Si_2$ and $CeCu_2Ge_2$, the second superconducting dome (on the right hand side of the diagram) is found to be on the border of a density transition characterized by a first order transition line and low temperature critical end point (red dot) followed by crossover behaviour (red dashed line). This is analogous to a liquid–gas transition. (The two domes overlap in the pure compounds but clearly separate in chemically doped specimens (red domes).)

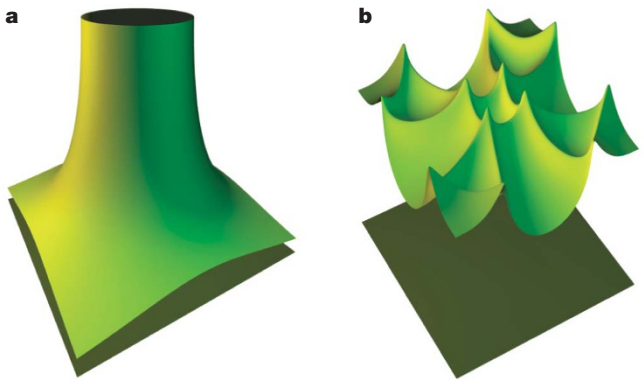


Figure 6 | Effect of crystal lattice on the magnetic properties. a, **b**, Static linear magnetic response (Lindhard function) versus wavevector in two dimensions for non-interacting electrons in a uniform background of neutralizing positive charges ('jellium'; **a**) and in the presence of a model crystal potential in a square lattice (**b**). Note the richer structure in the latter case. The different peaks give rise to oscillations of different wavelengths in real space, which can interfere constructively or destructively and thus lead to complex forms of the real-space magnetic interaction. As the Lindhard function is determined by the Fermi surface, it is as distinctive a property of a metal as is the Fermi surface itself.

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is, *p*-wave pairing does not lead to an efficient lowering of the total electronic energy[50,51]. These and other factors discussed elsewhere shed light on why the search for a superconducting analogue of liquid ^{3}He has proved so elusive[27,50,51].

In the light of the experimental and theoretical findings discussed above, we can establish a number of conditions that are particularly favourable for magnetic pairing. On the basis of the cases studied so far, one class of promising candidate material should have, at minimum, the following properties: (1) a single band of relatively high characteristic energy scale; (2) strong quasi-two-dimensional antiferromagnetic correlations for spin-singlet pairing and for large amplitude oscillations of the spin–spin interaction; and (3) a crystal structure that enables the repulsive regions of the pairing potential to be optimally neutralized. The quasi-two-dimensional organic superconductors[28] appear to go quite far in satisfying these conditions.

Although the applicability of the magnetic interaction model to the copper oxides[71] has been disputed, it is nevertheless interesting that the model provides a natural explanation for the robustness of superconductivity and correctly anticipated the *d*-wave symmetry[34] of the Cooper-pair state in this class of materials[72–75], which appear to optimally satisfy all of the above conditions.

In the framework presented in this Review, the antiferromagnetic correlations come about through the Lindhard function—that is, the response of the non-interacting itinerant electron system to a space- and time-dependent applied magnetic field—but, crucially, as enhanced by an appropriate effective (molecular) field. This approach is most appropriate to the description of delocalized quasiparticles not too close to a metal–insulator transition. Another approach considers localized models of magnetism in which antiferromagnetic correlations can arise and also lead to *d*-wave superconductivity (see refs 76–80 for recent discussions). These two approaches are normally associated with itinerant versus localized models of magnetism, and are most naturally suited to the 'over-doped' and 'under-doped' states of the copper oxides, respectively. The detailed understanding of the similarities and differences of these approaches to magnetically mediated superconductivity would be a major step forward.

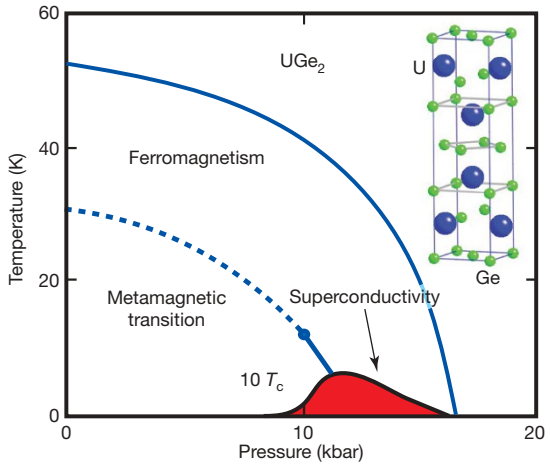


Figure 7 | Proximity to a ferromagnetic instability. Temperature–pressure phase diagram of the ferromagnetic superconductor UGe_2 (refs 27, 64, 65). The upper blue line corresponds to a ferromagnetic instability (analogous to the antiferromagnetic instability in Fig. 4) and the lower blue line to a metamagnetic instability with low critical end point (analogous to the density transition in Fig. 4). In this system, there is only one superconducting dome on the border of metamagnetism (a sudden change in the magnitude of the magnetization and density). A second superconducting dome is not observed on the border of the ferromagnetic transition. The strong first order nature of this transition, in contrast to the metamagnetic transition, leads one to expect relatively weak magnetic fluctuations and hence a weak magnetic pairing interaction.

The possibilities seem endless

In electrodynamics, the dimensionless coupling constant, which is a measure of the relative strength of the interaction, is 1/137 and hence small, leading only to minor corrections to the polarizer–analyser framework of the Coulomb interaction described earlier. The corresponding relevant dimensionless coupling constants characterizing the strength of the induced interaction (equation (2)) are likely to be much larger and possibly of order unity in real materials of interest here. It would therefore not be surprising if the lowest order interaction (equation (2)) fails to capture some essential physics. What is surprising is that it has led to so many useful insights and provided guidance in the searches for novel phenomena, some of which have proved to be fruitful.

The model interaction (equation (2)) was derived on the assumption that the characteristic amplitudes of the molecular fields are small. However, this is an unnecessary restriction, as the average over a gaussian distributed molecular field can now be carried out numerically by well-established Monte Carlo sampling techniques. One finds that when the characteristic dimensionless coupling constant is of order unity or greater, the model interaction (equation (2)) indeed fails to capture essential physics[81,82].

The quasiparticle-pairing problem where equation (2) ceases to be valid has not been studied in detail. At the level of approximation represented by equation (2), there is no fundamental difference in the various interaction channels (for example, density versus magnetic) in that, given the right conditions, they could all lead to robust forms of pairing. An understanding of the conditions favourable for high-temperature superconductivity cannot be obtained without considering the corrections to the simple theory[81,82]. This raises the question of whether one particular channel of interaction may be singled out by these higher corrections. In the preliminary non-perturbative (Monte Carlo) calculations, one finds that the magnetic interaction is enhanced by the higher order corrections, while the density interaction is reduced at least for the range of parameters considered. In this context, we note that in quantum electrodynamics the analogous corrections lead to an enhancement of the magnetic moment of the electron and therefore an enhancement of its magnetic coupling to the Maxwell field. The smallness of the enhancement in this latter case is due to the very weak coupling of the Maxwell field to the electron, in contrast to the case of the magnetic molecular field for which the corresponding enhancement can be substantial. These and previously mentioned features point to the special nature of spin in quantum mechanics, and may distinguish and favour the magnetic interaction over other interaction channels in the strong coupling limit.

The above non-perturbative treatment represents only one of many possible extensions of the model discussed in this Review for superconductivity without phonons. The richness of behaviour at even the simplest level of approximation points to the utmost importance of tuning material properties, and leaves no doubt that many more surprises are in store.

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